G	Clean shell	*C*
H	Replace tube bundle	*F, G*
I	Prepare shell pressure test	*D, E, H*
J	Prepare tube pressure test and reassemble	*I*

Draw a network diagram of activities for the project.

(*c*) A project has fourteen activities A through M. The relationships which obtain among these activities are given here. Construct the network and number them :

A is the first operation

B and *C* can be performed in parallel and are immediate successors to *A*

D, *E* and *F* follow *B*

G follows *E*

H follows *D*, but it cannot start until *E* is complete

I and *J* succeed *G*

F and *J* precede *K*

H and *I* precede *L*

M succeeds *L* and *K*

The last operation *N* succeeds *M* and *C*.

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Roll No. .....

576311(76)
676311(76)

MBA (3rd Semester)
Examination, Nov.-Dec., 2016

(New Scheme)

Optimization Methods

Time Allowed : 3 hours *Maximum Marks :* 80
Minimum Pass Marks : 32

Note : (*i*) There are **five** units.
(*ii*) Each unit consists of **three** questions (*a*), (*b*) and (*c*).
(*iii*) Attempt any **two** questions from each unit.
(*iv*) **All** questions carry equal marks.

UNIT - I

(*a*) How does the simplex algorithm indicate that–

(*i*) the problem has unbounded optimal solution;

(*ii*) the problem has no feasible solution.

(*b*) Use the graphical method to solve the following LP problem :

Maximize $Z = 2x_1 + 3x_2$

subject to the constraints $x_1 + x_2 \leq 30$

$$x_2 \geq 3$$

$$0 \leq x_2 \leq 12$$

$$0 \leq x_1 \leq 20$$

$$x_1 - x_2 \geq 0$$

and $x_1, x_2 \geq 0$

(*c*) Use penalty method to solve the following LP problem.

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

subject to the constraints

$$x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

and $x_1, x_2, x_3, x_4 \geq 10$

UNIT - II

(*a*) Solve the following transportation problem to minimize the total cost :

	Destinations			
Origins	D_1	D_2	D_3	Availability
Q_1	40	70	90	300
Q_2	12	80	30	400
Q_3	60	90	45	200
Requirement	300	300	300	

(*b*) A company receives a particular component required for its production from two suppliers A and B. It has production plants at three locations X, Y and Z. Due to non-availability of proper roads, the plant Y cannot receive the material from supplier A. Similarly, plant X cannot receive material from supplier B. The weekly requirements of the plants and weekly availability of supply from A and B along with the respective per unit cost of transportation is given below :

	X	Y	Z	Availability
A	9	—	5	45
B	—	7	6	35
Requirement	50	25	25	

Suggest an optimal purchase plan for the company so as to minimize the total transportation cost.

(*c*) A product is produced by four factories A, B, C and D. The unit production costs in them are Rs. 2, Rs. 3, Re. 1 and Rs. 5 respectively. Their production capacities are ; factory A–50 units, B–70 units, C–30 units and D–50 units. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transport cost in rupees from each factory to each store is given in the table below:

Stores

Factories	1	2	3	4
A	2	4	6	11
B	10	8	7	5
C	13	3	9	12
D	4	6	8	3

Determine the extent of deliveries from each of the factories of each of the stores so that the total production and transportation cost is minimum.

UNIT - III

(*a*) A small garment making unit has five tailors stitching five different types of garments. All the five tailors are capable of stitching all the five types of garments. The output per day per tailor and the profit (Rs.) for each type of garment are given below :

Tailors	Garments 1	2	3	4	5
A	7	9	4	8	6
B	4	9	5	7	8
C	8	5	2	9	8
D	6	5	8	10	10
E	7	8	10	9	9
Profit (Rs) per garment	2	3	2	3	4

If tailor D is absent for a specified period and no other substitute tailor is available, what should be the optimal assignment?

(*b*) A company has four territories open and four salesmen available for assignment. The territories are not equally rich in their sales potential. It is estimated that a typical salesman operating in each territory would bring in the following annual sales :

Territory	:	I	II	III	IV
Annual Sales (Rs.)	:	60,000	50,000	40,000	30,000

The four salesmen are also considered to differ in ability; it is estimated that working under the same conditions, their yearly sales would be proportionately as follows :

Salesman	:	A	B	C	D
Proportion	:	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is to assign the best salesman to the richest territory, the next best salesman to the second richest territory and so on. Verify this answer by the assignment method.

(*c*) What is the travelling salesman problem? Which situations can be treated as the travelling salesman problem? How does its solution differ from the solution of the assignment problem?

UNIT - IV

(*a*) Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call is assumed to be distributed exponentially, with mean 3 minutes. Find–

(*i*) the probability that an arrivals finds that four persons are waiting for their turn,

(*ii*) the average number of persons waiting and making telephone calls and

(*iii*) the average length of the queue that is formed from time to time.

(*b*) A self-service store employs one cashier at its counter. Nine customers are arrive on an average every 5 minutes while the cashier can serve 10 customers in 5 minutes. Assuming Poisson distribution for arrival rate and exponential distribution for service time. Find :

(*i*) Average number of customers in the system

(*ii*) Average number of customers in the queue

(*iii*) Average time a customer spends in the system

(*iv*) Average time a customer waits before being served

(*c*) A branch of Punjab National Bank has only one typist. Since the typing work varies in length (number of pages to be typed), the typing rate is randomly distributed approximating a Poisson

distribution with mean service rate of 8 letters per hour. The letters arrive at a rate of 5 per hour during the entire 8 hour-work day. If the typewriter is valued at Rs. 1.50 per hour, determine :

(*i*) Equipment utilization

(*ii*) The per cent time that an arriving letters has to wait

(*iii*) Average system time

(*iv*) Average cost due to waiting on the part of typewriter, i.e. it remain idle.

UNIT - V

(*a*) Define float. Explain its different types and their importances.

(*b*) Listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery :

Activity	Description	Predecessor Activity
A	Dismantle pipe connections	—
B	Dismantle heater, closure, and floating front	*A*
C	Remove tube bundle	*B*
D	Clean bolts	*B*
E	Clean heater and floating head front	*B*
F	Clean tube bundle	*C*